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prime gaps_COPPOLA_CIRO_ROME_ITALY – Version1.0 - November 2026

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The author used AI-assisted editing tools during manuscript preparation. All hypotheses, dataset construction, transition analysis and final scientific claims remain the responsibility of the author.

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Transition Graphs of Consecutive Prime Gaps: Preliminary Evidence for Forbidden Gap Transitions

By Ciro Coppola

Abstract

We introduce a graph-theoretic representation of consecutive prime gaps. Let $g_n = p_{n+1} - p_n$be the sequence of prime gaps. Instead of studying individual gaps, we consider transitions $(g_n, g_{n+1})$and represent them as directed edges in a transition graph.

Using a dataset containing 65,535 consecutive prime numbers from 2 to 821,603, we construct the corresponding transition matrix and identify a set of missing transitions. Some of these missing transitions are shown to correspond to inadmissible prime constellations under modular constraints. We introduce the notion of a forbidden transition and discuss its relationship with admissibility conditions from the theory of prime constellations. The work proposes a framework for studying local structural constraints in the sequence of prime gaps.

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1. Introduction

Prime gaps have been extensively studied within analytic and computational number theory. Traditionally, research focuses on:

- the distribution of primes p_n;
- the distribution of prime gaps g_n;
- admissible prime constellations;
- asymptotic properties of prime gaps.

In this work, we adopt a different perspective. Rather than considering individual gaps, we analyze transitions between consecutive gaps.

Given

$$g_n = p_{n+1} - p_n,$$

we study the ordered pairs

$$(g_n, g_{n+1}).$$

This naturally gives rise to a directed graph structure.

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2. Definitions

Definition 2.1 (Prime Gap)

Let

$$p_1, p_2, p_3, \dots$$

denote the increasing sequence of prime numbers.

The n-th prime gap is

$$g_n = p_{n+1} - p_n.$$

Definition 2.2 (Gap Transition)

A gap transition is an ordered pair

$$(g_n, g_{n+1}).$$

For example,

$$(2, 4)$$

denotes a transition from a gap of size 2 to a gap of size 4.

Definition 2.3 (Transition Graph)

The transition graph is the directed graph

$$G = (V, E)$$

where:

- $V$is the set of observed gap values;
- $(a, b) \in E$if the transition $a \to b$occurs at least once.

Definition 2.4 (Missing Transition)

A pair

$$(a, b)$$

is called a missing transition if it does not occur in the dataset.

Definition 2.5 (Forbidden Transition)

A missing transition is called forbidden if the associated prime configuration can be proven arithmetically impossible.

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3. Dataset

The study was conducted on a dataset containing:

Property	Value
Number of primes	65,535
Smallest prime	2
Largest prime	821,603

The dataset was extracted from .

The sequence was verified to contain no duplicates, no composite numbers and no missing primes within the covered interval.

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4. Gap Statistics

The observed gap distribution is dominated by small even gaps.

Gap	Frequency
6	11,428
4	6,932
2	6,906
12	6,683
10	5,976
8	4,715
18	4,058
14	3,554

The largest observed gap in the dataset is 114.

The average gap is approximately 12.54 and the median gap is 10.

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5. Transition Matrix

A transition matrix

$$M(a, b)$$

is constructed by counting occurrences of

$$a \to b.$$

Examples of verified transitions are:

Transition	Frequency
2→18	524
4→18	442
6→18	576

Transition	Frequency
8→18	281

Examples of missing transitions are:

$$2 \to 8,$$
$$2 \to 14,$$
$$2 \to 20,$$
$$4 \to 10,$$
$$4 \to 16,$$
$$4 \to 22,$$
$$8 \to 2,$$
$$8 \to 8,$$

6. Prime Configurations Associated with Transitions To a transition \[a\to b

we associate the configuration

$$\mathcal{C}(a,b) = \{0, a, a+b\}.$$

A realization of the transition corresponds to three consecutive primes

$$p, p+a, p+a+b.$$

The admissibility of the transition can therefore be studied through the admissibility of the associated configuration.

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7. Forbidden Transitions

Proposition 7.1

The transition

$$2 \to 8$$

is impossible.

Proof

The transition would require three consecutive primes:

$$p, p+2, p+10.$$

Modulo 3, the offsets are

$$\{0, 2, 1\},$$

which occupy all residue classes modulo 3.

Therefore one of the three terms must be divisible by 3.

The only possible exceptional case is $p = 3$, yielding

$$3{,}5{,}13.$$

However, 7 and 11 lie between 5 and 13, implying that the numbers are not consecutive primes.

Hence the transition cannot occur.

Proposition 7.2

The transition

$$4 \to 10$$

is impossible.

Proof

The corresponding configuration is

$$p, p+4, p+14.$$

The offsets modulo 3 are

$$\{0, 1, 2\},$$

which again occupy all residue classes modulo 3.

One of the terms must therefore be divisible by 3.

The exceptional case $p = 3$gives

$$3,7,17,$$

which are not consecutive primes.

Hence the transition cannot occur.

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8. Discussion

The results suggest that missing transitions can arise from at least two distinct causes:

1. arithmetic impossibility;
2. finite-sample absence.

This distinction motivates the introduction of forbidden transitions as a separate category.

The framework naturally connects transition graphs with classical admissibility conditions in the theory of prime constellations.

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9. Open Problems

Problem 1

Characterize all forbidden transitions.

Problem 2

Determine whether each missing transition is either:

- forbidden;

- statistically rare.

Problem 3

Study asymptotic properties of the transition graph as the prime range increases.

Problem 4

Investigate weighted versions of the graph and corresponding transition probabilities.

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10. Conclusion

We introduced a graph representation of consecutive prime-gap transitions and identified several missing transitions in a dataset of 65,535 consecutive primes.

Two of these transitions, namely

$$2 \to 8$$

and

$$4 \to 10,$$

were shown to be arithmetically impossible.

The notion of forbidden transition provides a natural bridge between transition graphs and admissible prime constellations. The results suggest that local constraints within the sequence of prime gaps may be fruitfully studied through graph-theoretic methods.

Appendix A - Dataset

Full description of the file, statistics and gap distribution.

Appendix B - Transition Matrix Construction

Algorithm for generating the matrix $M(a, b)$.

Appendix C - Computational Verification

Procedures used to check observed frequencies, absences, and transitions.

References

1. H. Hardy, J. Littlewood, *Some Problems of Partitio Numerorum*.
2. T. Oliveira e Silva et al., work on prime gaps.
3. Standard references on admissible prime constellations.
4. OEIS A001223 (Prime gaps).

APPENDIX A - DATASET DESCRIPTION AND PRELIMINARY STATISTICS

A.1 Dataset

The experimental analysis was performed using a dataset containing 65,535 prime numbers ranging from 2 to 821,603.

The dataset was verified for consistency and completeness.

The following checks were performed:

- all numbers are prime;
- no duplicated values are present;
- no prime numbers are missing within the covered interval;
- the sequence is strictly increasing.

A.2 Prime Gap Definition

Let:

p(1), p(2), p(3), ..., p(n)

denote the increasing sequence of prime numbers.

The corresponding prime gap sequence is defined as:

g(n) = p(n+1) - p(n)

Example:

p(n)	p(n+1)	g(n)
2	3	1
3	5	2
5	7	2
7	11	4
11	13	2

The complete dataset generates 65,534 effective gaps.

A.3 Global Statistics

Statistic	Value
Number of primes	65,535

Statistic	Value
Smallest prime	2
Largest prime	821,603
Mean gap	12.537
Median gap	10
Maximum gap	114

The maximum observed gap occurs between:

492113 and 492227.

A.4 Most Frequent Gaps

Gap	Frequency
6	11,428
4	6,932
2	6,906
12	6,683
10	5,976
8	4,715
18	4,058
14	3,554
16	2,396
24	2,219

APPENDIX B - CONSTRUCTION OF THE TRANSITION MATRIX

B.1 Transition Definition

Let:

g(1), g(2), ..., g(m)

be the gap sequence.

A transition is the ordered pair:

(g(i), g(i+1))

for i = 1,...,m-1.

B.2 Transition Matrix

The transition matrix M is defined as:

M(a,b)

where:

M(a,b) = number of occurrences of the transition

a -> b

within the gap sequence.

B.3 Example

Suppose the gap sequence is:

2, 4, 2, 6, 4

The resulting transitions are:

2 -> 4

4 -> 2

2 -> 6

6 -> 4

The corresponding matrix entries are:

Transition	Count
2 -> 4	1
4 -> 2	1
2 -> 6	1
6 -> 4	1

B.4 Computational Procedure

Pseudo-code:

INPUT:

gap sequence g[1..m]

FOR i = 1 TO m-1

a = g[i]

b = g[i+1]

M[a,b] = M[a,b] + 1

END FOR

The computational complexity is O(m).

B.5 Transition Graph

The matrix naturally induces a directed graph.

Vertices:

V = set of observed gaps

Edges:

(a,b) belongs to E

if and only if

M(a,b) > 0

The edge weight is equal to the observed frequency.

B.6 Missing Transitions

A transition is called missing whenever:

M(a,b) = 0

Examples observed in the dataset include:

- 2 -> 8
- 2 -> 14
- 2 -> 20
- 4 -> 10
- 4 -> 16
- 4 -> 22
- 8 -> 2
- 8 -> 8
- 8 -> 14

- 8 -> 20

APPENDIX C - COMPUTATIONAL VERIFICATION OF FORBIDDEN TRANSITIONS

C.1 Verification Method

The transition matrix was generated directly from the prime sequence.

All frequencies reported in this work were obtained from this matrix.

C.2 Verified Existing Transitions

The following transitions were confirmed to exist in the dataset:

Transition	Frequency
2 -> 18	524
4 -> 18	442
6 -> 18	576
8 -> 18	281

C.3 Verified Missing Transitions

The following transitions were confirmed to be absent:

Transition
2 -> 8
2 -> 14
2 -> 20
4 -> 10
4 -> 16
4 -> 22
8 -> 2
8 -> 8
8 -> 14

Transition
8 -> 20

C.4 Prime Configuration Associated with a Transition

For a transition:

a -> b

define the associated configuration:

C(a,b) = {0, a, a+b}

A realization of the transition corresponds to three consecutive primes:

p

p+a

p+a+b

C.5 Proven Forbidden Transitions

Proposition C.1

The transition:

2 -> 8

is impossible.

Associated configuration:

{0,2,10}

Modulo 3:

{0,2,1}

All residue classes modulo 3 are occupied.

After examination of the special case involving p = 3, the configuration is impossible.

Proposition C.2

The transition:

4 -> 10

is impossible.

Associated configuration:

{0,4,14}

Modulo 3:

{0,1,2}

All residue classes modulo 3 are occupied.

After examination of the special case involving p = 3, the configuration is impossible.

C.6 Open Cases

The following absent transitions remain theoretically unclassified:

- 2 -> 14
- 2 -> 20
- 4 -> 16
- 4 -> 22
- 8 -> 2
- 8 -> 8
- 8 -> 14
- 8 -> 20

Their absence has been experimentally verified, but a complete proof of impossibility has not yet been included in the present work.